



# Harsh Modi

Bachelor of Technology  
in Computer Science and Engineering  
Indian Institute Of Technology, Ropar

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## EDUCATION

| Degree                 | Institute/Board                       | CGPA/Percentage     | Year      |
|------------------------|---------------------------------------|---------------------|-----------|
| Bachelor of Technology | Indian Institute of Technology, Ropar | 8.13 (Till 4th Sem) | 2023-2027 |
| Senior Secondary       | Central Board of Secondary Education  | 95.6%               | 2023      |
| Secondary              | Central Board of Secondary Education  | 98.6%               | 2021      |

## EXPERIENCE

- Sifchain Finance** *Software Developer* April 2025 - Present  
Remote (USA)
  - Architected scalable AI applications using TypeScript and NLP models, contributing to Eliza OS open-source project and developing conversational AI agents with advanced reasoning capabilities.
  - Built React interfaces for agent visualization and implemented comprehensive testing frameworks (Jest, integration, end-to-end) across AI and web components.
  - Developed blockchain integration plugins for Polygon, Polygon zkEVM, and Polymarket, enabling AI agent interaction with DeFi protocols.
- Physics Wallah** *Full Stack Web Developer* April 2024 - August 2024  
Remote
  - Designed responsive UIs for Employee Management System using modern front-end technologies and integrated components with RESTful APIs, optimizing front-end, back-end, and database communication for enhanced performance.
  - Implemented geo-fencing technology for employee tracking and attendance, improving operational efficiency and data accuracy while ensuring intuitive user experience.

## PROJECTS

- Stock Price and Time Series Prediction** April 2024  
*RNNs, LSTM, Transformers, Hidden Markov Models* **GitHub**
  - Developed a comprehensive multi-model stock prediction system using LSTM, GRU, Transformer, and C-based HMMs with multi-head attention; integrated real-time APIs (Yahoo Finance, Alpha Vantage), intraday multi-timeframe analysis, sentiment models (VADER, TextBlob, FinBERT), market indices (S&P 500, NASDAQ, VIX, sector ETFs), and technical indicators (RSI, MACD, Bollinger Bands, ATR), achieving  $R^2$  of 0.94–0.98 and RMSE of 0.02–0.05 across large-cap tech stocks.
- CToPromelaAgentic: AI-Powered Formal Verification System** April 2025  
*Python, AI Agents, SPIN Model Checker, CIL, Streamlit, Machine Learning* **GitHub**
  - Architected an agentic AI-powered pipeline converting C/C++ to Promela via CIL IR, enabling formal verification of deadlock freedom, memory safety, and race conditions using SPIN; deployed autonomous agents with Google Gemini 2.0 for self-correcting error resolution, iterative code refinement, and ML-based pattern recognition across 15+ concurrency issues, featuring a Streamlit web interface and validation through 50+ real-world test cases for concurrency and protocol verification.
- Appointment Hub - Financial Services Platform** January 2024  
*Django, Python, SQL, Bootstrap, HTML/CSS, JavaScript* **GitHub**
  - Developed a full-stack appointment booking platform using Django with secure user authentication, role-based access control, and responsive dashboards for clients, financial professionals, and admins; implemented CSRF and XSS protection, secure password hashing, custom Vigenere cipher encryption, and a normalized relational database with UUID primary keys and optimized Django ORM queries for scalable and secure data management.
- Minor Projects**
  - RISC-V Simulator:** Built a pipelined C++ processor simulator in Python handling forwarding, hazard detection, and multi-stage execution; supports instruction-level debugging and performance metrics. **GitHub**
  - LLama3 Finetuning:** Fine-tuned Meta's LLaMA 3 model on custom domain-specific corpora using LoRA and HuggingFace's PEFT framework; enabled efficient inference and aligned responses for conversational agents. **GitHub**

## TECHNICAL SKILLS

- Programming Languages:** C/C++, Python, JavaScript, TypeScript
- Web Development:** HTML, CSS, JavaScript, Django, SQL, Bootstrap, React JS, Web Scraping, TypeScript, Web Sockets
- Machine Learning:** Supervised Learning, Unsupervised Learning, Ensembles, NLP, RNN, LSTM, Transformers, Transfer Learning, CNN, RAG, LLM, Embedding Models, BERT, Hugging Face, Agentic AI
- Blockchain:** Smart Contracts, ABIs, Ethers.js, Web3.js, Polygon, Polygon zkEVM, Polymarket

## KEY COURSES TAKEN

- CSE & Maths:** Algorithm & Data Structure, Digital Logic Design, Discrete Math, Calculus, Linear Algebra, Probability and Statistics, Programming Paradigms and Pragmatics, Computer Architecture
- Machine Learning:** Machine Learning Specialization, Natural Language Processing Specialization

## POSITIONS OF RESPONSIBILITY

- NLP Coordinator**, Iota Cluster, AI Club of IIT Ropar May 2024 - Current
- Secretary**, Enigma, Quizzing Club of IIT Ropar April 2025 - Current
- Event Management and Outreach Coordinator**, Enigma, Quizzing Club of IIT Ropar May 2024 - Current
- Event Co-Head**, Zeitgeist, Cultural Fest of IIT Ropar October 2024
- Workshop and Talks Co-Head**, Advitiya, Technical Fest of IIT Ropar February 2025

## ACHIEVEMENTS

- AIR 1388 (Top 0.1%), JEE Mains**, Secured All India Rank 1388 out of 1.2 million candidates in JEE Mains. 2023
- AIR 1614, JEE Advanced**, Ranked 1614 among 0.25 million candidates in one of the world's most challenging examinations. 2023
- Global Rank 7, Antigranular Hackathon**, Awarded €100 in a healthcare, data privacy and prediction hackathon;. 2024
- Winner (Rank 1), Predictathon - Advitiya**, Secured 1<sup>st</sup> position in a national AI/ML Hackathon; awarded 5000 and merchandise. 2025
- 2<sup>nd</sup> Runner-Up, ZenTej Hackathon**, Placed 3<sup>rd</sup> in an ML hackathon conducted by a Edify Japan. 2025
- Rank 9, Inter IIT Tech Meet 13.0**, Secured 9<sup>th</sup> position in the ISRO Problem Statement in the Inter IIT Tech Meet. 2024
- Expert, Codeforces**, Achieved 'Expert' rating on Codeforces competitive programming platform (ID: **FliccEatsBurger**) 2025